

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1533

Revere Dr 100 kV
October 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1533. The Interconnection Customer proposes a 162.5 MW battery energy storage facility interconnecting at the Revere Dr 100 kV substation in the Illinois area. The request was received on July 27, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 530 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$384,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Pueblo Xtra - Dairyman's Substation 230.0 kV Circuit 2
Contingency: Dairyman's - Pueblo Xtra Substation 230.0 kV
Loading: 103.7% (Rating: 1,013 MVA)
DFAX: 9.5%
Pre-project loading: 102.2%
2. Facility: Darlington County - Ceres 100.0 kV
Contingency: S Manheim - Zack 100.0 kV
Loading: 103.0% (Rating: 244 MVA)
DFAX: 5.1%
Pre-project loading: 99.6%
3. Facility: Pueblo Xtra - Dairyman's 230.0 kV Ckt 1
Contingency: Pueblo Xtra - Dairyman's 230.0 kV #2
Loading: 102.3% (Rating: 1,027 MVA)
DFAX: 9.5%
Pre-project loading: 100.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pueblo Xtra - Dairymans Substation 230.0 kV Circuit 2	Reconductor line	\$14,756,000	1% (\$160,000)
Darlington County - Ceres 100.0 kV	Reconductor line	\$548,000	4% (\$21,000)
Pueblo Xtra - Dairymans 230.0 kV Ckt 1	Reconductor line	\$18,683,000	1% (\$203,000)
	Total allocated to project		\$384,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.